

MEC6602 : Homework 1,
15% of final grade
Due date as per course Syllabus

Maximum 5 pages and can use appendices if you want to provide details that are secondary but whose understanding is important to your analysis. You will be judged on both content (75%) and form (25%). Content refers to the exercises requested, the precision of the analytical and numerical calculations (25 points), and the quality of the analysis of your results. **You are expected to implement the algorithm presented in class. Any deviation from these algorithms should be justified.** Form refers to style, clarity of presentation and quality of figures/tables. The detailed evaluation grid is:

Evaluation table

	points
exercises	5
precision	5
analysis	5
form	5

IMPORTANT: no points deducted for 'handwritten' or 'software editor' submissions, as the many equations are difficult and time-consuming to write on a computer! Put your efforts elsewhere!

Wave equation

The 1D wave equation is :

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

1. Perform a theoretical analysis of the accuracy, stability and consistency of the following algorithms:

a) Hybrid explicit-implicit with $\theta=0,0.5$ (Crank – Nicolson), 1.0

$$u_j^{n+1} + \theta \left(\frac{CFL}{2} \right) (u_{j+1}^{n+1} - u_{j-1}^{n+1}) = u_j^n - (1 - \theta) \frac{CFL}{2} (u_{j+1}^n - u_{j-1}^n)$$

b) Your own algorithm with 2nd order space, 4th order time. (It is not necessary to use a Runge-Kutta scheme to achieve this)

c) Your own algorithm with 4th order space, 2nd order time.

2. Write a program in the programming language of your choice to solve the above wave equation on the domain $[0, \pi]$ with these initial conditions :

$$u=0; x<0.5 \wedge x>1.0;$$

$$u=1; 0.5 \leq x \leq 1.0.$$

For each of the following algorithm, run numerical simulations varying the CFL number and report your observations when the mid-wave is at $x=2.5$. Comment on accuracy and stability.

a) Explicit 'backward', and 'forward'

$$\text{Backward : } u_j^{n+1} = u_j^n - \frac{c\Delta t}{\Delta x} (u_j^n - u_{j-1}^n)$$

$$\text{Forward : } u_j^{n+1} = u_j^n - \frac{c\Delta t}{\Delta x} (u_{j+1}^n - u_j^n)$$

$$\text{c) Leap-Frog : } u_j^{n+1} = u_j^{n-1} - \frac{c\Delta t}{\Delta x} (u_{j+1}^n - u_{j-1}^n)$$

$$\text{d) Lax-Wendroff: } u_j^{n+1} = u_j^n - \frac{c\Delta t}{2\Delta x} (u_{j+1}^n - u_{j-1}^n) + \frac{1}{2} \left(\frac{c\Delta t}{\Delta x} \right)^2 (u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

$$\text{e) Lax: } u_j^{n+1} = (u_{j+1}^n + u_{j-1}^n) / 2 - \frac{c\Delta t}{2\Delta x} (u_{j+1}^n - u_{j-1}^n)$$

d) Hybrid explicit-implicit with $\theta=0, 0.5$ (Crank - Nicolson), 1.0

$$u_j^{n+1} + \theta \left(\frac{CFL}{2} \right) (u_{j+1}^{n+1} - u_{j-1}^{n+1}) = u_j^n - (1 - \theta) \frac{CFL}{2} (u_{j+1}^n - u_{j-1}^n)$$

e) Your own algorithm with 2nd order space, 4th order time.

f) Your own algorithm with 4th order space, 2nd order time.

Note the following theoretical result for the 1D wave equation:

	Stability	Consistency	Spatial Order	Time Order
Explicit Backward	CFL lower than 1	Yes	1	1
Explicit Forward	Unstable	Yes	1	1
Lax	CFL lower than 1	No	1	2
Lax-Wendroff	CFL lower than 1	Yes	2	2
Leap-Frog	Neutrally stable	Yes	2	2

Steady 1D Euler equations

The quasi 1-D Euler equations are, in conservative form

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = S$$

with

$$Q = \begin{bmatrix} \rho A \\ \rho u A \\ e A \end{bmatrix}, F = \begin{bmatrix} \rho u A \\ (\rho u^2 + P) A \\ u(e + P) A \end{bmatrix}, S = \begin{bmatrix} 0 \\ P \frac{dA}{dx} \\ 0 \end{bmatrix}$$

closed by the perfect gas law: $P = (\gamma - 1) \left(e - \frac{\rho u^2}{2} \right)$. The symbol definitions are in standard textbooks.

Implement the MacCormack and the implicit Beam-Warming algorithm

Apply this model to the flow in a nozzle with an area distribution

$$A(x) = 1.398 + 0.347 \tanh(0.8x - 4) \text{ for } 0 < x < 10$$

The Mach at the inlet is 1.25. For this stationary problem, perform a numerical simulation and analysis using the two algorithms above. Study the following cases:

- 1) Supersonic inlet and supersonic outlet;
- 2) Supersonic inlet and subsonic outlet with a back pressure 1.9 times the inlet pressure.